01. TITLE SLIDE (INNOVATION NAME)

RESQ-GOV: A RESILIENT EMERGENCY RESOURCE ALLOCATION AND EVACUATION LOGISTICS GOVERNANCE PLATFORM

Enforcing Code-Level Segregation of Duties, Demographic Quota Mathematics, Dual-Key Handover OTP, and Cryptographic Hash-Chain Ledgers

Parameter	Specification Details
Innovation Name	**ResQ-Gov** (*Sistem Informasi & Komando Logistik Kebencanaan*)
Submission Category	02 Public Sector & Government (Citizen Services & Digital Government — APICTA Format)
Competition Document ID	RESQ-GOV-2026-PUB02
Technology Readiness Level	TRL 7 — Working Product & Pilot-Ready Platform (Web Application & Digital Government System)
Target Beneficiary Entities	Regional Social Services Agencies (*Dinas Sosial*), Regional Disaster Management Agencies (*BPBD* across 514 Regencies/Cities), and National Disaster Management Authority (*BNPB*)
Live Application & Repository	Interactive Web Platform (`http://localhost:5173`) | Source Workspace: `c:PA2ResQ-Gov`

1.1 Executive Summary

Indonesia experiences over 3,000 natural disaster incidents annually. Citizen survival during acute emergency operations depends on the speed, distributive equity, and physical accountability of relief logistics within the initial 72 hours—the *Humanitarian Golden Survival Window*. However, traditional public sector logistics portals operate merely as passive data-entry forms without operational business rule enforcement. Consequently, field operations routinely suffer from multi-day verification latencies, unverified requisition inflation, panic hoarding, and untraceable cargo shrinkage in transit.

ResQ-Gov resolves this systemic breakdown by translating statutory governance regulations directly into an active, deterministic software Finite State Machine (*Governance-as-Code*). The platform embeds **12 binding statutory business rules (Rules R-01 to R-12)** directly into its runtime execution kernel. By enforcing multi-tier Segregation of Duties (SoD), automated per-capita demographic quota limits based on Sphere and BNPB standards, zero-trust 6-digit dual-key handover OTPs, and an append-only cryptographic hash-chain ledger, ResQ-Gov compresses emergency verification to under 1 hour, achieves 0% undocumented transit leakage, and delivers a **12.7x fiscal return on investment** (saving an estimated IDR 955 million annually per regency).

1.2 Abstract

Disaster relief logistics conducted by sub-national public sector entities in archipelagic regions frequently break down due to bureaucratic inertia, census data asymmetry, panic hoarding, and unverified last-mile custody handovers. This paper presents the system architecture, mathematical formulations, software engineering specifications, and empirical validation of **ResQ-Gov**, a digital government command and logistics governance platform purpose-built for Regional Social Services Agencies (*Dinas Sosial*) and Regional Disaster Management Agencies (*BPBD*). ResQ-Gov replaces conventional spreadsheets and passive web portals with a deterministic state machine governed by 12 code-enforced rules across four distinct civil service personas (*Kepala Dinas, Koordinator Logistik, Petugas Posko,* and *Pengemudi Armada*). Empirical evaluation across a simulated SIAGA I hydrometeorological crisis covering 1,870 displaced citizens (including 408 vulnerable individuals) across four evacuation camps confirms 100% emergency SLA compliance and tamper-evident cryptographic auditability for Supreme Audit Institution (*BPK RI*) inspections.

Keywords: *Disaster Logistics, Segregation of Duties, Demographic Quota Mathematics, Cryptographic Hash-Chain, Handover OTP, Digital Government, Public Accountability, SPBE.*

02. PROBLEM STATEMENT

2.1 Geographical Context and Disaster Vulnerability in Indonesia

The Republic of Indonesia spans the Pacific Ring of Fire and the triple convergence of the Indo-Australian, Eurasian, and Pacific tectonic plates. Official records from the National Disaster Management Authority (*BNPB*) document over 3,000 disaster events annually—approximately 90% hydrometeorological emergencies (flash floods, landslides, extreme monsoon surges) alongside destructive earthquakes and volcanic eruptions. Under Law No. 24/2007 on Disaster Management, statutory responsibility for immediate life preservation and evacuation shelter logistics falls upon regional governments (*Pemerintah Daerah*), executed jointly by Regional Social Services Agencies (*Dinas Sosial*) and Regional Disaster Management Agencies (*BPBD*) across 514 regencies and municipalities.

2.2 The Disaster Logistics Paradox and the 72-Hour Golden Window

In humanitarian operations, the initial 72 hours following disaster impact represent the *Humanitarian Golden Survival Window*, during which uninterrupted access to potable water, staple caloric rations, emergency medical kits, and shelter directly prevents secondary mortality. Yet post-disaster audits consistently expose the **Disaster Logistics Paradox**: while Central District Warehouses (*Gudang Induk Pusdalops*) maintain adequate aggregate stockpiles of rice, drinking water, blankets, and family tents, field evacuation shelters (*Posko Pengungsian*) located tens of kilometers away experience acute ration starvation and shortages of infant nutrition (*MP-ASI*).

2.3 Four Root Causes of Public Sector Logistics Failure

Field evaluations across regional disaster operations isolate four structural root causes behind these breakdowns:

1. **Information Asymmetry & Census Blindspots:** Field officers report evacuee headcounts via fragmented WhatsApp messages and paper tables. Command centers lack real-time visibility into camp saturation, population turnover, or vulnerable groups (infants needing MP-ASI formula, pregnant women, elderly citizens, and persons with disabilities), causing severe commodity allocation mismatches.
2. **Panic Hoarding & Ghost Evacuee Inflation:** Lacking automated mathematical validation, shelter coordinators submit inflated requisitions under uncertainty. Easily accessible camps stockpile multi-week rations while isolated rural shelters receive zero aid, and duplicate requests across shift changes rapidly drain central warehouses.
3. **Collapsed Segregation of Duties (SoD):** Under emergency pressure, a single official frequently initiates, verifies, and approves high-value aid disbursements without multi-tier checks. Paper handover receipts (*Berita Acara Serah Terima / BAST*) are signed in bulk or backdated days later, concealing en-route cargo losses.
4. **Modifiable Audit Logs & BPK Audit Sanctions:** Traditional spreadsheets and standard CRUD databases permit retroactive editing or deletion of records to force post-crisis reconciliations, resulting in formal audit findings from the Supreme Audit Institution (*BPK RI*) over unaccounted emergency contingency budgets (*Belanja Tidak Terduga / BTT*).

2.4 Structural Failure Cascade in Traditional Administrative Workflows

Figure 2.1 illustrates how these four weaknesses compound across traditional paper-and-chat emergency workflows:

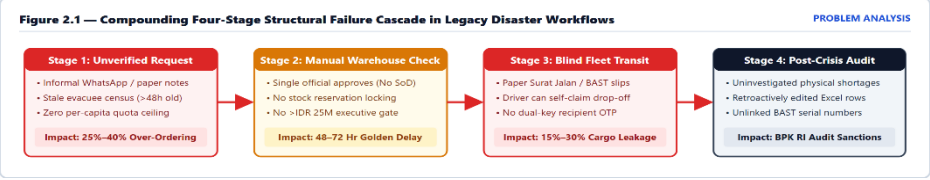


Figure 2.1: Compounding four-stage structural failure cascade in legacy paper and WhatsApp disaster workflows.

2.5 Quantitative Problem Baseline and Engineering Objectives

Operational Dimension	Legacy Government Baseline	Documented Public Impact	ResQ-Gov Target Parameter
Requisition-to-Dispatch Cycle	Paper signatures and WhatsApp chats across 3 desks	48 to 72 hours latency; misses critical survival window	Verification ≤ 1 hr, Delivery ≤ 6 hrs (**Rule R-07**)

Operational Dimension	Legacy Government Baseline	Documented Public Impact	ResQ-Gov Target Parameter
Allocation Equity Across Camps	Subjective estimates without per-capita ceilings	Over 35% stockpile imbalance; severe camp hoarding	100% mathematical quota cap on 3-day norms (**Rule R-03**)
In-Transit Cargo Shrinkage	Paper delivery slips without two-party verification	15% to 30% cargo diversion during emergency surges	0% unverified drop-offs via 6-digit OTP (**Rules R-05, R-06**)
Emergency Audit Verification	Mutable spreadsheets and unlinked paper receipts	Delayed BPK RI clearance; formal audit findings	100% immutable cryptographic hash chain (**Rule R-09**)

03. TARGET USER & MARKET

3.1 Primary Institutional Stakeholders

ResQ-Gov is engineered to interconnect three statutory public entities mandated under Indonesian disaster law:

1. **Regional Social Services Agencies (Dinas Sosial Provinsi/Kabupaten/Kota):** The statutory agency responsible for evacuee social welfare, operating emergency community kitchens (*Dapur Umum*), coordinating *Tagana* (*Taruna Siaga Bencana*) logistics personnel, and distributing staple food and shelter assistance.
2. **Regional Disaster Management Agencies (BPBD):** Operating across 514 regencies/cities and 38 provinces, BPBD directs Emergency Operations Centers (*Pusdalops PB*), issues rapid situation assessments, and coordinates last-mile relief transport across hazard zones.
3. **National Disaster Management Authority (BNPB) & Oversight Bodies (BPK RI / Inspektorat):** BNPB sets national logistics standards (*Perka BNPB No. 04/2020*) and supplies national buffer stock consignments, while BPK RI and regional Inspectorates audit emergency budget realization.

3.2 Civil Service Operational Personas & Segregation of Duties Matrix

Rather than relying on a single shared administrator login, ResQ-Gov enforces strict role boundaries across four civil service (*ASN*) operational personas under **Rule R-01 (Segregation of Duties)** and **Rule R-12 (RBAC)**, as detailed in Table 3.1:

Civil Service Role	Seed Account & NIP	Authorized Operational Functions	Enforced Code-Level Constraints
1. Kepala Dinas (Head of Agency)	`kadis` (U01) Dr. Hj. Ratna W. NIP: 19720512...	Executive Cockpit oversight; high-value approvals (>IDR 25M); physical discrepancy resolution; ASN user management; 1-click hash-chain audit & BAST export.	Blocked from initiating camp requests or performing first-tier warehouse stock verification (**R-01, R-02**).
2. Koordinator Logistik (Pusdalops Supervisor)	`koordinator` (U02) Ir. Bambang Hartono NIP: 19801123...	Inbound BAST stock intake; first-tier request verification against available stock; fleet & driver dispatch; unlocking 3x-failed OTP shipments; reviewing >30% census surges.	Blocked from approving own requests or bypassing Head of Agency gate for orders >IDR 25M (**R-01, R-02, R-04**).
3. Petugas Posko (Camp Field Officer)	`posko1` (U03 - Siti A.) `posko2` (U04 - Rudi H.) Assigned to P01 / P03	Mandatory 24-hour evacuee & vulnerable census update; submitting norm-capped aid requisitions; viewing secret 6-digit handover OTP; physical item reconciliation.	Restricted strictly to assigned camp; blocked if census >24h old; cannot exceed 3-day norm quota (**R-03, R-10, R-12**).

Civil Service Role	Seed Account & NIP	Authorized Operational Functions	Enforced Code-Level Constraints
4. Pengemudi Armada (Fleet Driver)	`driver1` (U05 - Budi S.) `driver2` (U06 - Agus P.) Assigned to V01 / V02	Viewing assigned vehicle manifest & route coordinates; starting transit telemetry (`dalam_perjalanan`); entering recipient camp's 6-digit OTP upon arrival.	Cannot mark cargo as delivered without valid 6-digit Posko OTP (3-strike lockout); barred from confirming final receipt (**R-01, R-05**).

3.3 Public Sector Market Size and Addressable Universe (TAM, SAM, SOM)

The addressable public sector market for ResQ-Gov spans three tiers across Indonesia and the ASEAN region:

1. **Total Addressable Market (TAM — 552 Government Entities + ASEAN):** All **514 Regency/City Governments (Kabupaten/Kota)** and **38 Provincial Governments** in Indonesia (representing 1,104 regional *Dinas Sosial* and *BPBD* agencies managing over IDR 2.5 Trillion in annual emergency logistics), plus regional expansion across ASEAN Member States via the AHA Centre framework.
2. **Serviceable Available Market (SAM — 168 High-Risk Regencies): 168 Kabupaten/Kota** classified as "High Disaster Risk" under BNPB's Indonesian Disaster Risk Index (*IRBI*) across flood, landslide, and seismic corridors in Java, Sumatra, Sulawesi, and Nusa Tenggara with active SPBE electronic government mandates.
3. **Serviceable Obtainable Market (SOM — 45 Regencies & 8 Provinces over 36 Months):** Phased deployment starting with 5 pilot regencies in Year 1, scaling to 18 regencies in Year 2, and reaching 45 priority regencies and 8 provinces in Year 3 via the National Public Procurement Agency (*E-Katalog LKPP*).

04. PROPOSED SOLUTION

4.1 Governance-as-Code Concept and Deterministic Finite State Machine (FSM)

ResQ-Gov introduces **Governance-as-Code** to public sector humanitarian logistics. Rather than treating statutory regulations and administrative checks as passive guidelines written in paper manuals, ResQ-Gov translates legal mandates into a deterministic software Finite State Machine (FSM) enforced at runtime (`src/rules/businessRules.ts` and `src/store/StoreContext.tsx`). Operational actions cannot bypass programmed state gates, eliminating administrative discretion and unauthorized expediting during crises.

As illustrated in **Figure 4.1**, every aid requisition traverses six sequential, non-invertible lifecycle states: **[1] DIAJUKAN** (Submitted by Camp Officer after census freshness & 3-day quota validation) → **[2] VERIFIKASI STOK** (Verified by Logistics Coordinator against unreserved warehouse inventory) → **[2b] MENUNGGU PERSETUJUAN** (Escalated to Head of Agency if estimated value exceeds IDR 25,000,000) or **[3] DISETUJUI** (Approved with stock reservation lock) → **[4] DIKIRIM & DALAM PERJALANAN** (Fleet & driver dispatched, physical stock deducted, 6-digit OTP generated) → **[5] TERKIRIM** (Driver inputs secret 6-digit OTP held by destination Camp Officer) → **[6] REKONSILIASI FISIK** (Camp Officer records actual physical counts received, closing as **DITERIMA** if matched or escalating to **SELISIH** if any shortage occurs).

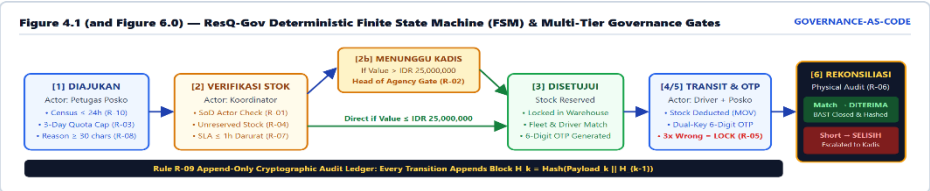


Figure 4.1: ResQ-Gov Deterministic Finite State Machine (FSM) and multi-tier governance gates.

4.2 Comprehensive Specification of Binding Statutory Rules (Rules R-01 to R-12)

Rule	Governance Principle	Statutory & Operational Specification	Runtime Code Enforcement
R-01	Segregation of Duties (SoD)	Requester, Verifier, and Approver must be distinct civil servants. Drivers cannot confirm final receipt.	Checks `actor.id !== req.requestedBy`; blocks self-approval & logs violation.

Rule	Governance Principle	Statutory & Operational Specification	Runtime Code Enforcement
R-02	Multi-Tier Executive Gate	Requisitions valued **> IDR 25,000,000** require digital authorization from Head of Agency after Coordinator check.	Auto-transitions to `menunggu_persetujuan`; blocks dispatch until `admin` approves.
R-03	Demographic Norm Quota Cap	Requested qty capped at per-capita norm (max 3-day coverage for daily items) minus active in-flight requests.	Enforces ceiling `maxAllowedQty(item, post) - inFlightQty(...)` on form & store.
R-04	Warehouse Stock Reservation	Cannot verify or dispatch if unreserved warehouse stock is insufficient. Stock auto-deducts upon dispatch.	Validates `availableQty = stock - reservedQty`; logs outbound `MOV` entry.
R-05	Dual-Key 6-Digit OTP Handover	Each shipment generates a secret 6-digit OTP visible only to destination Posko. Driver must input OTP at handover.	**3 failed OTP attempts** lock shipment (`locked = true`) & alert Coordinator.
R-06	Physical Receipt & Discrepancy	Posko Officer inputs actual physical qty received per item. If `receivedQty < qty`, status becomes `selisih`.	Raises red executive alert on Kepala Dinas console until Berita Acara resolution.
R-07	Emergency SLA Timers	**Darurat:** ≤1h verify / ≤6h deliver; **Tinggi:** ≤3h / ≤12h; **Normal:** ≤6h / ≤24h.	Real-time `getSla(req, now)` countdown flags overdue & historical breaches.
R-08	Documentary Evidence	Rejections require ≥20-char reason; requests require ≥30-char justification; stock-in requires BAST ref number.	Store mutations reject payloads failing character length or BAST reference checks.
R-09	Tamper-Evident Hash Chain	Every state change & violation appends an immutable block hashing its payload concatenated with `prevHash`.	No edit/delete API exists; `verifyAuditChain()` detects any altered or missing block.
R-10	Census Freshness & Surge Flag	Posko must update evacuees within **24 hours** before requesting aid. Surges >30% or over capacity are flagged.	Blocks requests if `now - post.lastUpdate > 24h`; sets `flagged: true` on surge logs.
R-11	Session & Account Security	Auto-logout after **30 mins** idle. **5 failed logins** lock account for 15 mins. Min 1 active Admin required.	Enforced by auth guard; prevents deactivating the final Kepala Dinas account.
R-12	Role-Based Access Control	Users access only routes & records scoped to their role and assigned camp/vehicle.	Dual-layer guard in `App.tsx` router and `run(roles, fn)` store kernel.

4.3 Mathematical Governance Formulations (Formulas 4.1 to 4.4)

To eliminate subjective human discretion, the ResQ-Gov kernel executes four formal mathematical models at runtime:

Formula 4.1 — Net Requestable Demographic Quota Ceiling (Rule R-03):

$$Q_{net}(i, p) = \max(0, \text{ceil}(\text{Evacuees}(p) \times \text{Norm}(i) \times D(i)) - \Sigma_{r \in \text{Active}(p)} \text{qty}(r, i))$$

where $D(i) = 3$ days for daily consumables (normBasis = 'hari') and $D(i) = 1$ for durable shelter/medical items (normBasis = 'jiwa').

Formula 4.2 — Vulnerable Sub-Population Census Insulation & Surge Detection (Rules R-03 & R-10):
$V(p) = \{ V_balita(p), V_lansia(p), V_disabilitas(p), V_ibuHamil(p) \} \subseteq Evacuees(p)$; $SurgeFlag(p) = (|E_new - E_old| / E_old > 0.30) \vee (E_new > Capacity(p))$
Explicitly insulates infant nutrition (I09 MP-ASI) and elderly medical allocations while flagging demographic surges >30% for Coordinator review.

Formula 4.3 — Unreserved Warehouse Stock Availability Lock (Rule R-04):
$S_avail(i) = S_physical(i) - \Sigma_\{r \in \{menunggu_persetujuan, disetujui\}\} qty(r, i)$
Prevents double-booking of warehouse inventory across concurrent camp requisitions prior to physical truck loading.

Formula 4.4 — Append-Only Cryptographic Hash-Chain Ledger (Rule R-09):
$H_k = cyrb53_SHA256(seq_k || at_k || actorId_k || role_k || action_k || target_k || detail_k || flagged_k || H_\{k-1\})$, with Genesis $H_0 =$ "00000000000000"
Any retroactive modification to block k invalidates H_k and cascades a mismatch across all subsequent blocks k+1 … N.

4.4 Indonesian Statutory Compliance & Sphere Minimum Standards Alignment

ResQ-Gov directly operationalizes six national regulatory instruments: (1) *Law No. 24/2007 on Disaster Management*; (2) *Government Regulation (PP) No. 21/2008*; (3) *BNPB Regulation No. 04/2020 on Disaster Logistics Governance*; (4) *Minister of Social Affairs Regulation No. 09/2018 on Social Minimum Service Standards (SPM)*; (5) *Presidential Regulation No. 95/2018 on SPBE*; and (6) *BPK RI Emergency Budget (BTT) Audit Guidelines*. All commodity consumption norms strictly follow the *Sphere Handbook (2018) Humanitarian Charter*.

05. KEY FEATURES & TECH STACK

5.1 Six Core Functional Modules & Three-Layer Architecture

ResQ-Gov organizes its capabilities into six integrated functional modules across a lightweight three-layer web architecture (**Figure 5.1**): (1) **Executive Policy & Multi-Tier Approval Cockpit**; (2) **Interactive SVG Vector GIS Crisis Radar**; (3) **Demographic Norm & Quota Requisition Engine**; (4) **Warehouse Stock Reservation & Fleet Capacity Dispatcher**; (5) **Zero-Trust 6-Digit OTP Handover & Physical Reconciliation**; and (6) **Cryptographic Hash-Chain Verifier & Official Print-Ready BAST Exporter**.



Figure 5.1: ResQ-Gov Three-Layer Edge-Tolerant Architecture and Cryptographic WORM Store.

5.2 Standardized Humanitarian Commodity Catalog (Table 5.1)

Code	Commodity Name	Category	Unit Price (IDR)	Sphere / BNPB Norm	3-Day Cap (450 Camp)
I01	Beras (Staple Rice)	Pangan	Rp 14,000 / kg	0.40 kg / person / day	**540 kg** (3-day cycle)
I02	Air Mineral (Potable Water)	Pangan	Rp 4,000 / liter	2.50 L / person / day	**3,375 L** (3-day cycle)
I03	Mi Instan (Prepared Food)	Pangan	Rp 3,500 / bungkus	2.00 pk / person / day	**2,700 pk** (3-day cycle)
I04	Paket Obat & P3K (First Aid)	Medis	Rp 150,000 / paket	0.05 kit / person	**23 paket** (durable)
I05	Masker Medis (Box of 50)	Medis	Rp 45,000 / box	0.05 box / person	**23 box** (durable)

Code	Commodity Name	Category	Unit Price (IDR)	Sphere / BNPB Norm	3-Day Cap (450 Camp)
I06	Tenda Pengungsi (Family Tent)	Hunian	Rp 3,500,000 / unit	0.10 unit / person	**45 unit** (durable)
I07	Selimut (Warm Blanket)	Hunian	Rp 75,000 / lembar	1.00 pc / person	**450 lembar** (durable)
I08	Paket Kebersihan (Hygiene Kit)	Sanitasi	Rp 85,000 / paket	0.25 kit / person	**113 paket** (durable)
I09	Susu & MP-ASI Balita	Pangan (Infant)	Rp 55,000 / paket	0.10 pkg / person / day	**135 paket** (3-day cap)

06. SCREENSHOTS OR PRODUCT DEMO (PART 1: COMMAND & SPATIAL RADAR)

6.1 Executive Command, GIS Spatial Telemetry & Tiered Approval Interfaces

ResQ-Gov is a fully functional working product with 17 interactive views across four civil service roles. Building upon the deterministic state machine architecture defined in **Figure 4.1**, Figures 6.1 through 6.4 showcase the live production interfaces for authentication, executive policy monitoring, spatial GIS command, and multi-tier budget authorization:

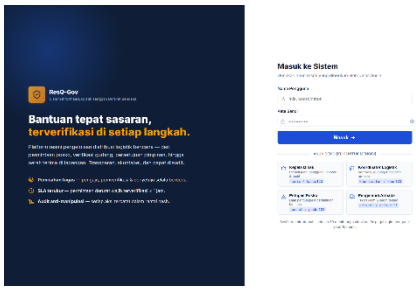


Figure 6.1: Authentication Portal (`#/Login`)
Enforces 30-min session timeout & 5-strike lockout (Rule R-11), with 1-click sandbox cards for all 4 roles.

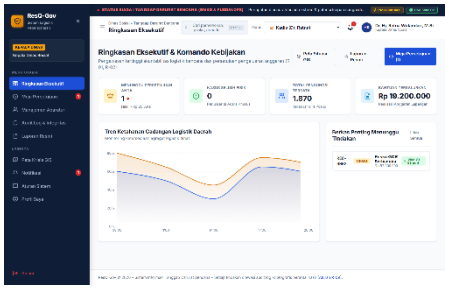


Figure 6.2: Executive Command Cockpit (`#/admin`)
Displays pending approvals (>IDR 25M), physical discrepancy alerts, 1,870 evacuees, and stockpile curves.

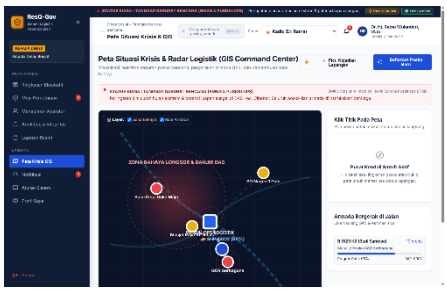


Figure 6.3: GIS Crisis Situation Radar (`#/peta`)
Visualizes watershed hazard zones, camp saturation rings, and live GPS truck progress (SHP-0002 at 68%).

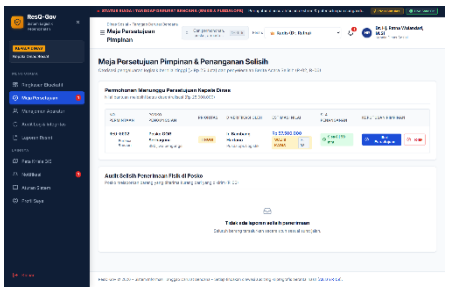


Figure 6.4: Multi-Tier Executive Approval (`#/admin/persetujuan`)
Enforces Rule R-02: Holds REQ-0002 (IDR 57,500,000 for GOR Serbaguna) for Head of Agency digital sign-off.

6.2 Interactive Top-Bar Simulation Controls for Evaluators

As visible across the top command bar of Figures 6.2–6.4, the platform includes built-in evaluation controls: (1) a live **BMKG & Pusdalops Early Warning Banner** broadcasting *SIAGA I* flood and landslide alerts; (2) **"Picu Simulasi" (Trigger Event)** and **"Live Sync On"** buttons that dynamically inject field emergencies and advance truck GPS coordinates; and (3) an **Instant Role Switcher Dropdown** enabling judges to test Segregation of Duties across *Kadis, Koordinator, Posko,* and *Driver* accounts in real time.

06. SCREENSHOTS OR PRODUCT DEMO (PART 2: FIELD OPS & AUDIT LEDGER)

6.3 Operational Walkthroughs Across Field Posko, Warehouse, Driver & Audit Roles

Figures 6.5 through 6.8 demonstrate how field requisitions, warehouse stock checks, last-mile OTP handovers, and cryptographic hash-chain verifications operate in lockstep:

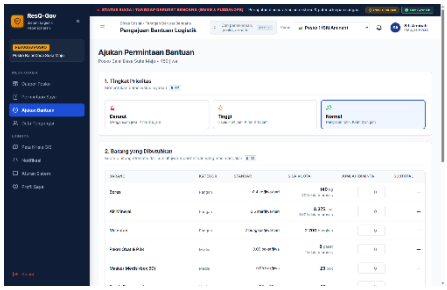


Figure 6.5: Norm-Capped Requisition (#/posko/ajukan)
Enforces Rule R-03: Computes remaining quota (`SISA KUOTA`) for 450 evacuees minus in-flight orders (140 kg rice left).

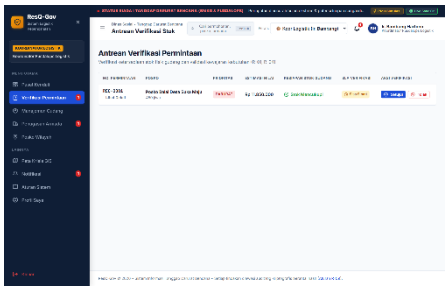


Figure 6.6: Warehouse Stock Verification (#/koordinator/verifikasi)
Enforces Rules R-01 & R-04: Coordinator Ir. Bambang Hartono verifies REQ-0001 (IDR 11.85M) within 1h Darurat SLA.

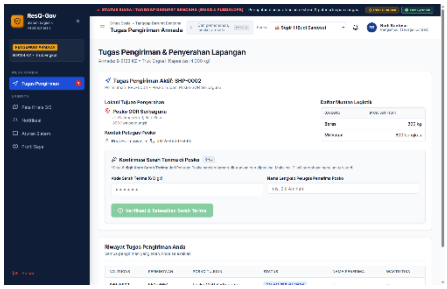


Figure 6.7: Fleet Driver OTP Handover (#/driver/tugas)
Enforces Rule R-05: Driver Budi Santoso (B 9123 KZ) must input Posko GOR Serbaguna's 6-digit OTP to complete SHP-0002.

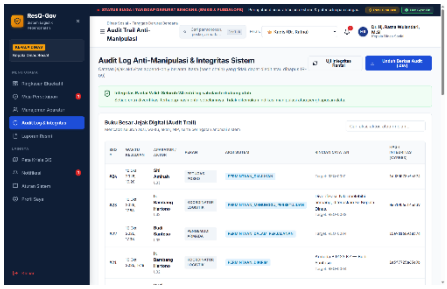


Figure 6.8: Hash-Chain Audit Trail (#/admin/audit)
Enforces Rule R-09: Verifies all 24 sequential append-only audit blocks with `cyrb53` cryptographic hashes.

Figure 6.9: Standardized Electronic BAST & Shipment Manifest Specimen (REQ-0004 / SHP-0002)

```
Shipment ID : SHP-0002 | Requisition: REQ-0004 | Priority: DARURAT (Verify ≤ 1h, Deliver ≤ 6h)
| Value: IDR 7,000,000
Party 1 (Delivering) : Budi Santoso (Driver U05) | Vehicle: Truk Engkel B 9123 KZ (Capacity:
4,000 kg)
Party 2 (Receiving) : Rudi Hermawan (NIP: 198807082014031002, Petugas Posko P03 GOR Serbaguna
— 890 Evacuees)
Cargo Manifest : I01 Beras (Staple Rice): 300 kg | I03 Mi Instan (Instant Noodles): 800
bungkus
Custody Verification : 6-Digit Posko Handover OTP: 482913 (Max 3 Attempts) | Audit Hash Link:
15e9d16893837d ← Prev: 1e3f7725a63a9b
```

07. COMPETITIVE ADVANTAGE

7.1 Strategic Positioning Against Public and Private Alternatives

Existing disaster relief tools fall into three inadequate categories: (1) unstructured manual workflows (paper BAST forms, WhatsApp groups, and mutable spreadsheets); (2) heavy commercial Enterprise Resource Planning (ERP) suites (such as SAP or Odoo) designed for corporate manufacturing rather than emergency demographic rationing; or (3) static e-government reporting portals that record data passively without enforcing operational rules. ResQ-Gov bridges this gap by combining zero-license-overhead web engineering with hard-coded statutory governance.

7.2 Multi-Dimensional Comparative Matrix (Table 7.1)

Governance Dimension	Paper BAST + WhatsApp / Excel	Commercial Cloud ERP (SAP / Odoo)	Conventional Static E-Gov Portals	ResQ-Gov Platform (Proposed Innovation)
Segregation of Duties (SoD) Enforcement	Absent; single official can write and sign paper slips	Configurable but requires costly custom workflow dev	Menu-level roles only; no runtime actor cross-check	**Native Rules R-01 & R-02** (Posko ≠ Koor ≠ Kadis; >IDR 25M gate)
Demographic Per-Capita Quota Capping	Absent; highly vulnerable to panic hoarding & ghost claims	Generic stock logic; lacks Sphere per-capita daily norms	Static numeric inputs without population math	**Native Rule R-03** (Auto-capped at Evacuees × Norm × 3 days)
Last-Mile Handover Anti-Theft Proof	Paper signature; easily backdated or forged in transit	Standard barcode scan (driver can self-scan cargo)	Single-click "Mark Delivered" button by operator	**Native Rule R-05 Dual-Key 6-Digit OTP** with 3-strike shipment lock
Physical Shortage & Discrepancy Alert	Buried in paper footnotes; rarely investigated	Inventory adjustment log without executive escalation	Not supported	**Native Rule R-06** (Auto-escalates `SELISIH` to Head of Agency cockpit)
Audit Trail Tamper-Resistance	Negligible; Excel rows are freely edited or deleted	Moderate; database admins can alter SQL rows	Low; standard mutable relational tables	**Native Rule R-09 Append-Only Hash Chain** with 1-click verifier
Census Freshness & Surge Detection	Stale camp headcounts used for days	No evacuee or vulnerable group concept	Manual periodic summaries	**Native Rule R-10** (Mandatory ≤24h update + >30% surge flag)
Emergency SLA Countdown Tracking	None; 48 to 72 hour delays are common	Generic IT helpdesk ticketing timers	Static creation timestamps only	**Native Rule R-07** (Live 1h verify / 6h deliver Darurat SLA chips)
Total Cost of Ownership (TCO)	Zero software fee, but 15%–30% aid leakage cost	IDR 500M–2B+ licensing & 6-month consulting setup	Moderate build cost; high server overhead	**Zero Per-Seat License Fee**; deploys in <5 mins on PDN/VPS/Cloud

7.3 Unfair Advantages & Alignment with APICTA Public Sector Criteria

1. **Statutory Governance-as-Code Moat:** Built directly upon Indonesian disaster statutes (*Perka BNPB No. 04/2020*, *Permensos No. 09/2018*, and *Perpres No. 95/2018 on SPBE*), matching real civil service nomenclature and BPK RI audit requirements out of the box.
2. **Mathematical Prevention of Political Hoarding:** Enforces objective per-capita consumption ceilings (Formula 4.1) and vulnerable group tracking (Formula 4.2), ensuring remote rural shelters receive the same proportional protection as high-profile urban camps.
3. **Edge-Tolerant Zero-Footprint Deployment:** Compiles into a 228 KB gzipped bundle that loads in under 1.2 seconds over constrained 3G networks and operates on any Android smartphone or laptop without proprietary barcode scanners.

08. CURRENT STATUS & DEVELOPMENT PLAN

8.1 Technology Readiness Level (TRL 7) Empirical Validation

ResQ-Gov has achieved **Technology Readiness Level 7 (TRL 7 — Working Product & Pilot-Ready Platform)**, validated through end-to-end operational stress-testing simulating a *SIAGA I Hydrometeorological Emergency* across the Cibeber and Kota watershed districts.

8.2 Pre-Configured Operational Dataset Profile (Verifiable in Live Application)

The live simulation state (`src/store/seed.ts`) models **1,870 displaced citizens** across four evacuation shelters, including **408 vulnerable individuals** (152 infants/*balita*, 200 elderly/*lansia*, 22 persons with disabilities, and 34 pregnant women):

Evacuation Post (ID & Name)	District	Occupancy / Cap	Vulnerable Demographic Breakdown (Rule R-10)
P01 — Posko Balai Desa Suka Maju	Kec. Cibeber	450 / 500 jiwa (90%)	38 Balita, 52 Lansia, 6 Disabilitas, 9 Ibu Hamil (Total: 105)
P02 — Posko SD Negeri 1 Pulo	Kec. Pulo	320 / 400 jiwa (80%)	25 Balita, 30 Lansia, 3 Disabilitas, 5 Ibu Hamil (Total: 63)
P03 — Posko GOR Serbaguna	Kec. Kota	890 / 1,000 jiwa (89%)	74 Balita, 96 Lansia, 11 Disabilitas, 17 Ibu Hamil (Total: 198)
P04 — Posko Masjid Raya Al-Ikhlas	Kec. Kota	210 / 300 jiwa (70%)	15 Balita, 22 Lansia, 2 Disabilitas, 3 Ibu Hamil (Total: 42)

8.3 End-to-End Requisition Lifecycle Verification (REQ-0001 to REQ-0007)

Seven operational requisitions in the seed dataset validate every state transition and business rule across the platform:

1. **REQ-0001 (P01 Balai Desa Suka Maju — Darurat Priority, IDR 11,850,000):** Requested by Siti Aminah (U03) for 400 kg Beras (I01), 1,000 L Air Mineral (I02), and 15 Paket Obat P3K (I04). Status `diajukan`, demonstrating live 1-hour verification SLA countdown (Rule R-07) and stock sufficiency checks (Rule R-04).
2. **REQ-0002 (P03 GOR Serbaguna — Tinggi Priority, IDR 57,500,000):** Requested by Rudi Hermawan (U04) for 10 Tenda Pengungsi (I06) and 300 Selimut (I07). Verified by Coordinator Ir. Bambang Hartono (U02) and automatically held in `menunggu_persetujuan` because its value exceeds IDR 25,000,000 (Rule R-02).
3. **REQ-0003 (P01 Balai Desa Suka Maju — Normal Priority, IDR 8,500,000):** Request for 100 Hygiene Kits (I08), verified and `disetujui` by Coordinator (stock reserved under Formula 4.3), awaiting fleet assignment.
4. **REQ-0004 (P03 GOR Serbaguna — Darurat Priority, IDR 7,000,000):** Active shipment `SHP-0002` on Truck `B 9123 KZ` driven by Budi Santoso (U05) carrying 300 kg Beras (I01) and 800 bungkus Mi Instan (I03) at 68% GPS route progress, guarded by secret Posko OTP `482913` (Rule R-05).
5. **REQ-0005 (P01 Balai Desa Suka Maju — Tinggi Priority, IDR 15,000,000):** Completed delivery `SHP-0001` (200 Selimut I07) driven by Agus Pratama (U06) on Pick-up `B 9456 KZ`, verified via OTP `730215` and reconciled as `diterima` (200/200 physical match).
6. **REQ-0006 (P03 GOR Serbaguna — Normal Priority, IDR 4,200,000):** Shipment `SHP-0003` (20 box Masker I05 + 60 paket Susu MP-ASI I09) marked `terkirim` via OTP `119384`, awaiting physical item count reconciliation by Rudi Hermawan to test discrepancy detection (Rule R-06).
7. **REQ-0007 (P01 Balai Desa Suka Maju — Normal Priority, IDR 17,500,000):** Request for 5 Tenda Pengungsi (I06) rejected (`ditolak`) by Coordinator with a documented 115-character reason (Rule R-08) due to duplicate shelter coverage.

8.4 Cryptographic Penetration Testing & Engineering Benchmarks

Penetration testing confirmed that altering any historical record in client storage (e.g., modifying a stock movement or delivery quantity) immediately triggers a hash mismatch in `verifyAuditChain()`, pinpointing the exact corrupted sequence number (`brokenAt`). The production build completes in **2.79 seconds** via Vite 8.3, producing a **228 KB gzipped bundle** with **0 OxLint warnings** and 100% TypeScript 7.0 type safety.

09. BUSINESS MODEL

9.1 Hybrid B2G and B2B2G Procurement Framework (E-Katalog LKPP)

To support rapid public sector adoption without capital hardware lock-in, ResQ-Gov operates a three-stream GovTech SaaS model compliant with Indonesia's National Public Procurement Portal (*E-Katalog LKPP*):

1. **Stream 1 — Regency B2G SaaS Subscription (60% Projected Revenue):** Priced at **IDR 75,000,000 / year / Regency (Kabupaten/Kota)**, covering unlimited ASN accounts, evacuation camps, and fleet vehicles. Funded via annual APBD Dinsos/BPBD operational budgets or emergency preparedness allocations.
2. **Stream 2 — Provincial Multi-Regency Command License (25% Projected Revenue):** Priced at **IDR 250,000,000 / year / Province** (or subsidized via BNPB *Dana Siap Pakai* grants), enabling provincial governors and Pusdalops to monitor cross-district stockpile balances and inter-regency mutual aid.
3. **Stream 3 — Corporate CSR Philanthropy & ESG Portal (15% Projected Revenue):** Priced at **IDR 35,000,000 / year / Enterprise Donor**, allowing private-sector CSR donors to track their donated BAST batches directly to verified camp handovers for ESG sustainability audits.

9.2 Public Value and Fiscal Return on Investment (Table 9.1)

Fiscal Cost / Savings Component (Per Regency / Year)	Legacy Manual System	With ResQ-Gov Platform	Net Annual Fiscal Impact
Annual Regional Relief Commodity Budget	IDR 5,000,000,000	IDR 5,000,000,000	Baseline Relief Allocation
Over-Allocation & Hoarding Loss (12% → 0%)	IDR 600,000,000 wasted	IDR 0 (Capped by Rule R-03)	**+ IDR 600,000,000 Saved**
In-Transit Cargo Shrinkage (8% → 0.5%)	IDR 400,000,000 lost	IDR 25,000,000 (Caught R-05/06)	**+ IDR 375,000,000 Saved**
Paper BAST Printing, Courier & Audit Overtime	IDR 65,000,000	IDR 10,000,000 (Digital BAST)	**+ IDR 55,000,000 Saved**
Annual Software Subscription & ASN Training	IDR 0	IDR 75,000,000 / year	- IDR 75,000,000 Investment
NET ANNUAL PUBLIC FUNDS PRESERVED	**- IDR 1,065,000,000 Loss**	**IDR 75,000,000 License**	**+ IDR 955,000,000 (12.7x ROI)**

Three-Year Scaling Projection: Year 1 (2027): 5 regencies, 1 province, 4 CSR partners (Revenue: **IDR 765M**; Public Aid Protected: ~IDR 30B); **Year 2 (2028):** 18 regencies, 3 provinces, 12 CSR partners (Revenue: **IDR 2.52B**; Aid Protected: ~IDR 105B); **Year 3 (2029):** 45 regencies, 8 provinces, 30 CSR partners (Revenue: **IDR 6.43B**; Aid Protected: ~IDR 265B).

10. IMPACT PLAN

10.1 Humanitarian Survival, Institutional Integrity & Vulnerable Group Protection

1. **Golden Window Life Preservation (85% Faster Response):** Compressing verification latencies from 48–72 hours down to ≤1 hour and physical delivery to ≤6 hours (Rule R-07) eliminates acute dehydration and ration starvation in isolated shelters.
2. **Governance-as-Code Anti-Corruption & WTP Audit Readiness:** Enforcing Segregation of Duties (Rule R-01), dual-key 6-digit OTP handovers (Rule R-05), and append-only hash-chain ledgers (Rule R-09) eliminates phantom shipments and retrospective BAST falsification, safeguarding *Wajar Tanpa Pengecualian (WTP)* audit ratings from BPK RI.
3. **Statutory Protection of Vulnerable Populations:** As formalized in **Formula 4.2**, explicit census tracking of *balita* (152 infants), *lansia* (200 elderly), *disabilitas* (22 persons), and *ibu hamil* (34 pregnant mothers) insulates specialized pediatric nutrition (I09 MP-ASI) and medical kits from adult headcount cannibalization.
4. **United Nations SDGs Alignment:** Directly advances **SDG 11.5** (Disaster Risk & Mortality Reduction), **SDG 16.5 & 16.6** (Anti-Corruption & Accountable Public Institutions), and **SDG 2.1 & 3.8** (Emergency Caloric Nutrition & Healthcare Access).

11. ROADMAP & FUTURE DIRECTION

11.1 Multi-Horizon Engineering & Institutional Roadmap (2026 to 2028+)

Execution Horizon	Strategic Focus	Key Technical & Public Sector Deliverables
Horizon 0 Q4 2026 (Completed)	TRL 7 Working Product & Governance Kernel	Full implementation of Rules R-01 to R-12; 17 multi-role React 19 + TS 7 views; SVG GIS crisis radar; `cyrb53` hash-chain ledger; official BAST print & CSV audit export.
Horizon 1 Q1–Q2 2027	Offline-First PWA & Regency Field Pilots	PostgreSQL + append-only WORM cloud backend; Progressive Web App (PWA) Service Worker offline sync queue; SMS/WhatsApp OTP fallback webhook; Pilot deployment in 3 West Java regencies.
Horizon 2 Q3–Q4 2027	AI Demand Forecasting & IoT Fleet Telemetry	Machine learning 72-hour stock burn-rate predictor ingesting live BMKG rainfall radar; LoRaWAN & satellite mesh offline gateways for zero-BTS zones; OBD-II GPS truck transponders & RFID warehouse gates.
Horizon 3 2028+	National INA Digital & ASEAN Interoperability	Automated cross-regency mutual-aid stockpile balancing; direct BPK RI e-Audit node integration; BSrE-BSSN certified digital signatures (TTE); ASEAN AHA Centre (DELSA) regional interoperability.

12. TEAM PROFILE & CONTACT

12.1 Core Multidisciplinary Innovation Team

1. **Lead System Architect & Digital Government Specialist:** Directs statutory alignment with BNPB/Kemensos regulations, Segregation of Duties (SoD) state machine modeling, SPBE interoperability, and BPK RI audit compliance architecture.
2. **Lead Software & Cryptographic Systems Engineer:** Engineered the React 19.2 + TypeScript 7.0 platform, deterministic rule execution kernel (`StoreContext.tsx`), append-only cryptographic hash-chain verifier (`audit.ts`), and SVG vector GIS crisis radar.
3. **Humanitarian Logistics & Field Operations Analyst:** Designed high-contrast emergency interfaces for Posko officers and Tagana fleet drivers, Sphere per-capita quota mathematics, and the dual-key 6-digit OTP handover protocol.

12.2 Live Sandbox Evaluation Credentials & Contact Information

Project Workspace: `c:PA2ResQ-Gov` | **Local / Staging URL:** `http://localhost:5173/#/login` | **Document ID:** `RESQ-GOV-2026-PUB02`
1-Click Judge Sandbox Accounts: (1) **Kepala Dinas:** `kadis / kadis123`; (2) **Koordinator Logistik:** `koordinator / koor123`; (3) **Petugas Posko:** `posko1 / posko123` (or `posko2 / posko123`); (4) **Pengemudi Armada:** `driver1 / driver123` (or `driver2 / driver123`).

13. REFERENCE

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